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(54) **METHOD OF CLEANING FLUID DISPENSER BY APPLYING SUCTION FORCE AND VIBRATING MENISCUS**

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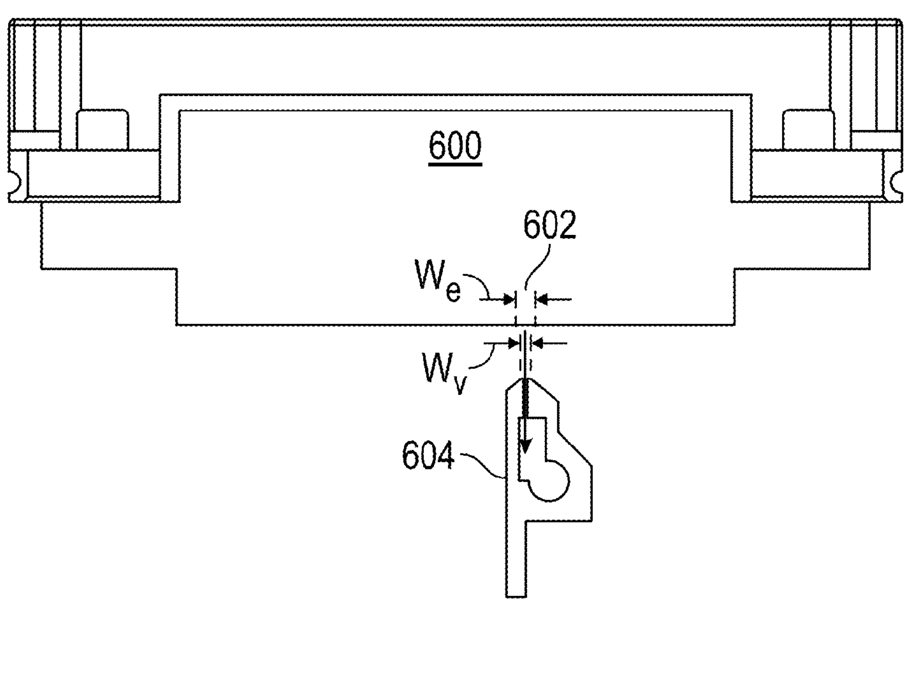
(57) **ABSTRACT**

A method of cleaning a fluid dispenser for dispensing a material during non-contact maintenance of the fluid dispenser. The fluid dispenser including a plurality of nozzles disposed on a faceplate. The method including applying a suction force onto a surface of the faceplate using a suction apparatus, the suction apparatus being translated from one end of the faceplate to an opposite end of the faceplate such that a portion of nozzles from the plurality of nozzles are exposed to the suction force. The method continues by vibrating a menisci of the portion of nozzles that are exposed to the suction force to remove at least a portion of the material accumulated on the faceplate.

16 Claims, 8 Drawing Sheets



(51) **Int. Cl.**

B08B 7/02	(2006.01)
B29C 64/112	(2017.01)
B29C 64/35	(2017.01)
B41J 2/045	(2006.01)
G03F 7/00	(2006.01)
G03F 7/16	(2006.01)

(52) **U.S. Cl.**

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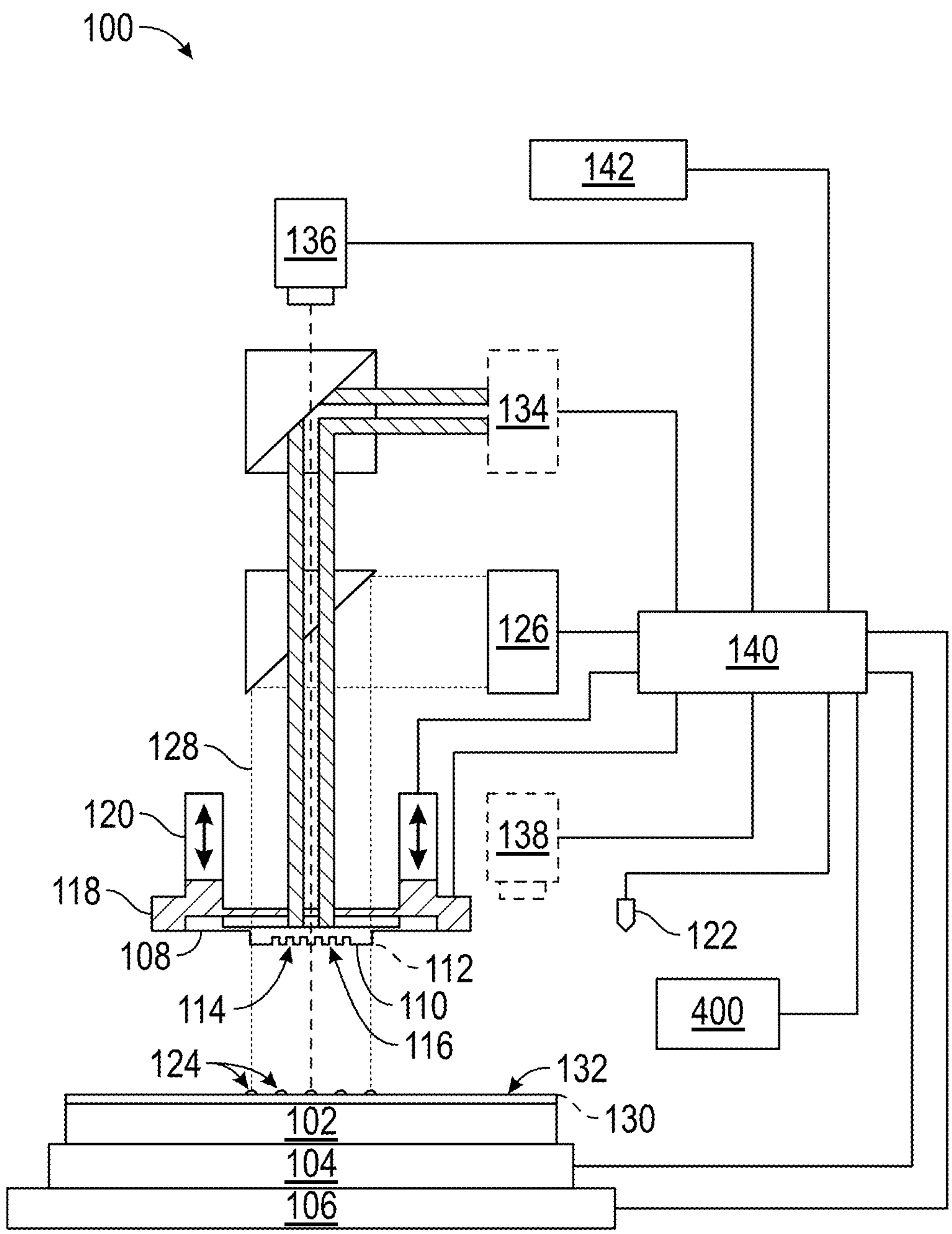
100
142
136
134
126
140
128
120
118
138
122
108
112
114
110
116
400
124
132
130
102
104
106

FIG. 1

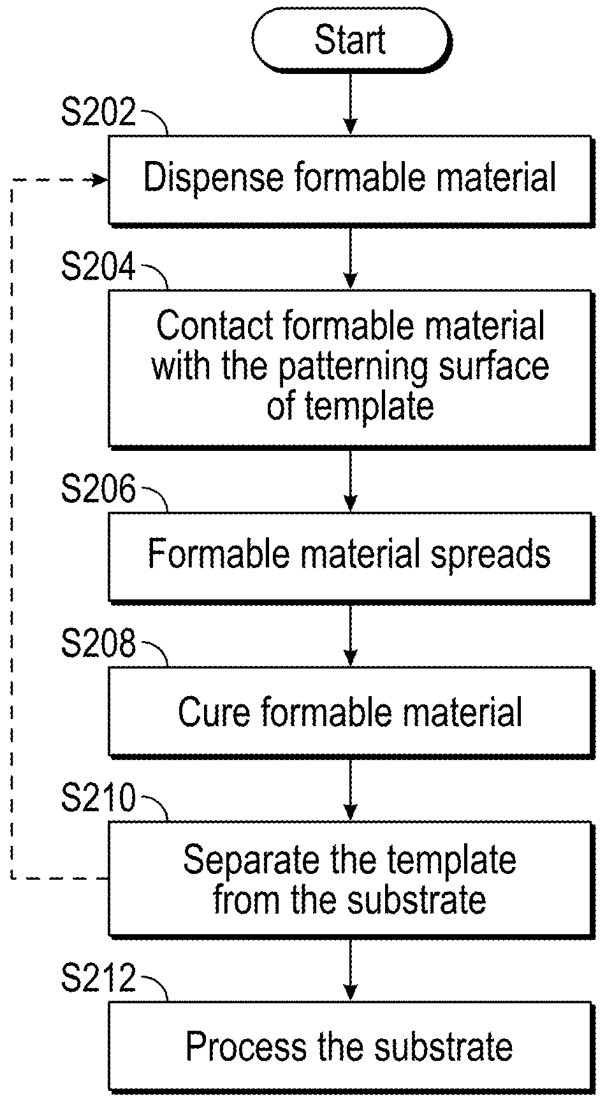
Start
S202
Dispense formable material
S204
Contact formable material with the patterning surface of template
S206
Formable material spreads
S208
Cure formable material
S210
Separate the template from the substrate
S212
Process the substrate

FIG. 2

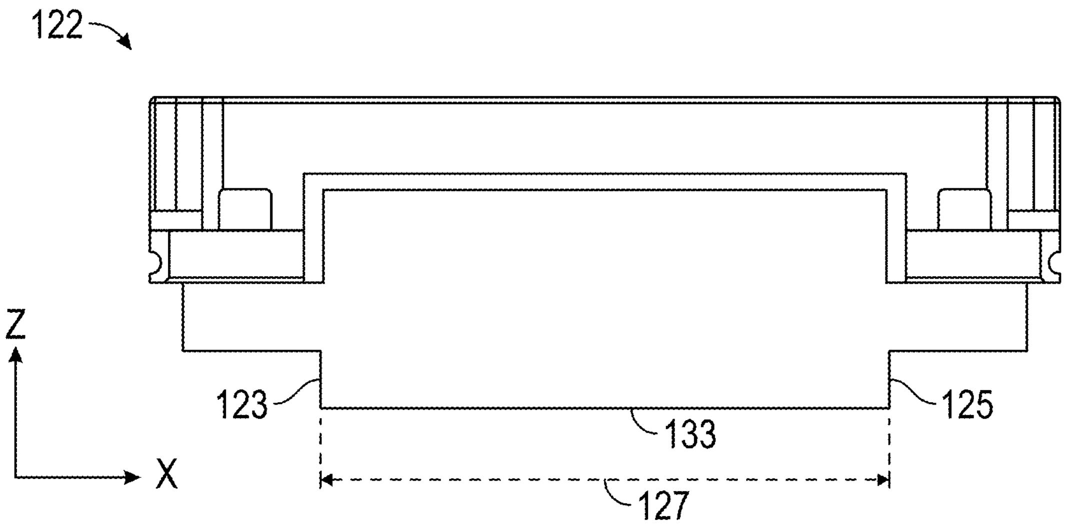

FIG. 3A

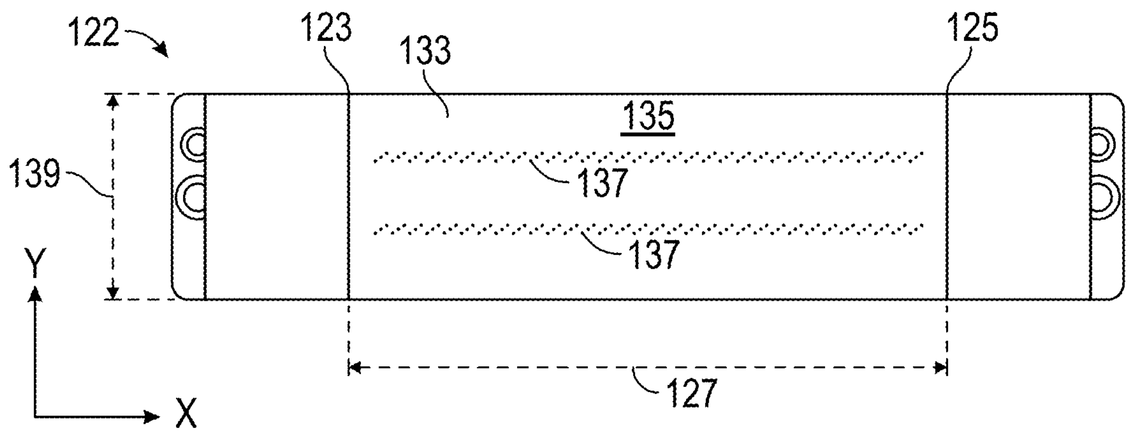

FIG. 3B

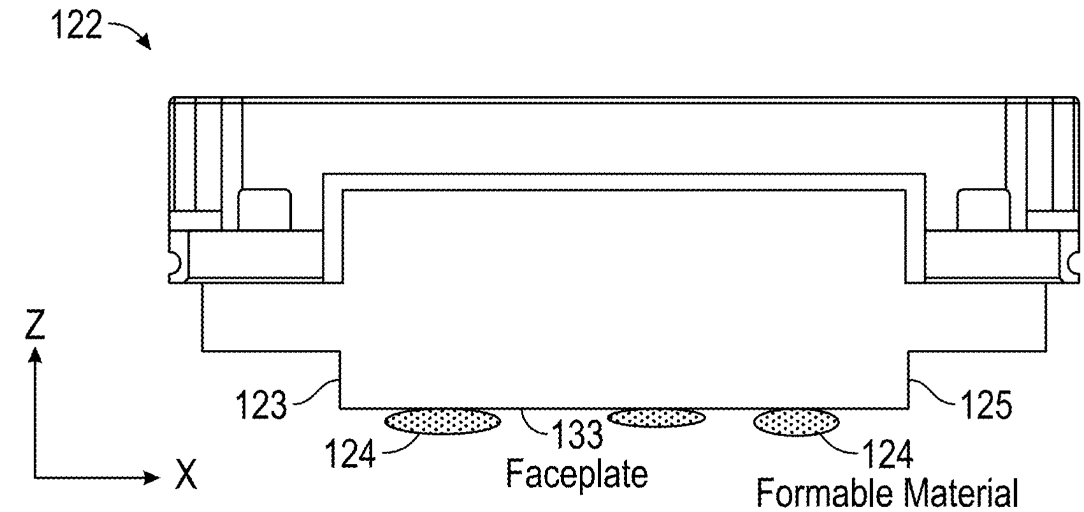
122
Z
X
123
125
124
133
Faceplate
124
Formable Material

FIG. 4A

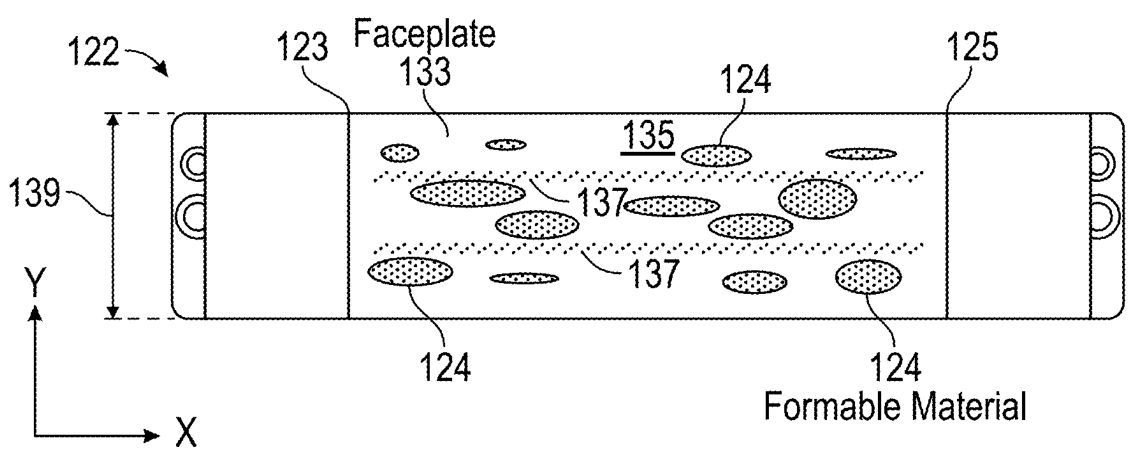
122
123
Faceplate
133
125
124
135
139
137
137
Y
X
124
124
Formable Material

FIG. 4B

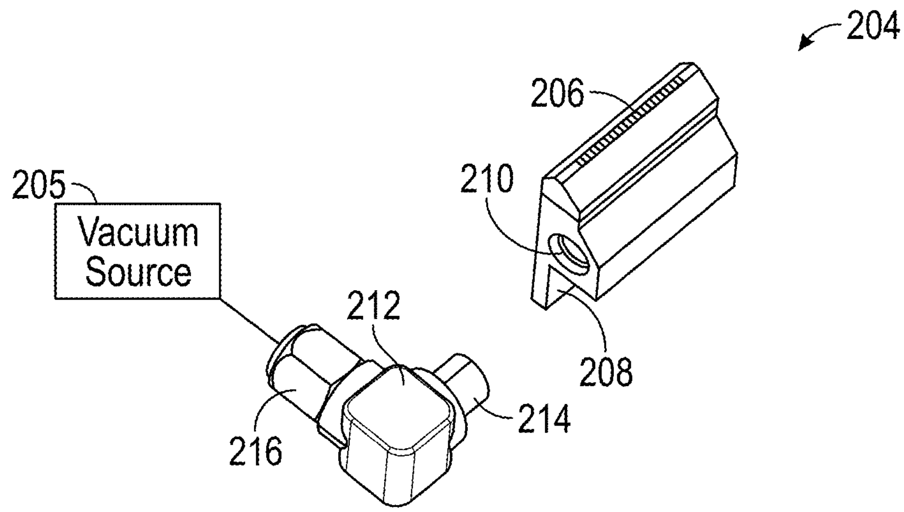
204
206
210
205
Vacuum Source
212
214
216
208

FIG. 5

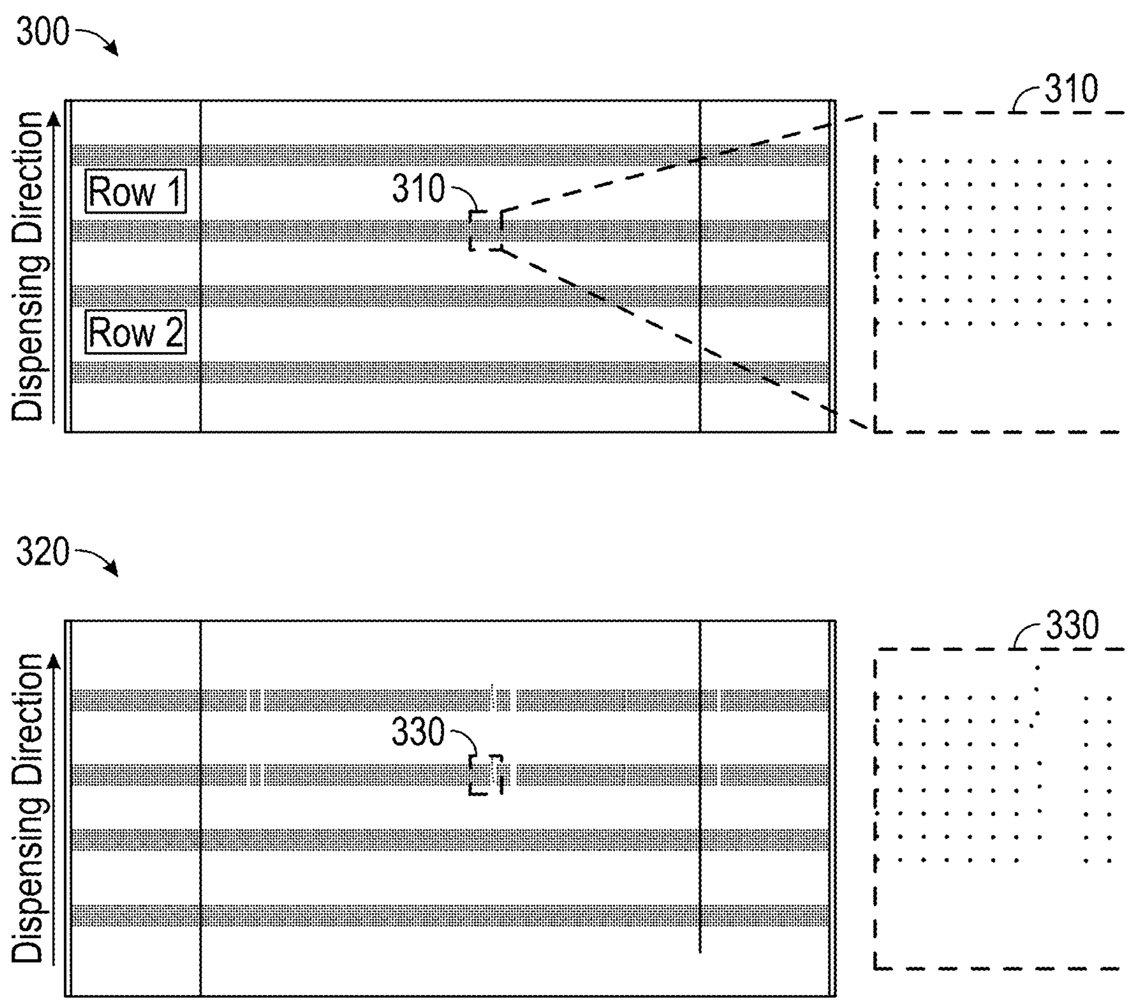
300
310
Dispensing Direction
Row 1
Row 2
310
320
330
Dispensing Direction
330

FIG. 6

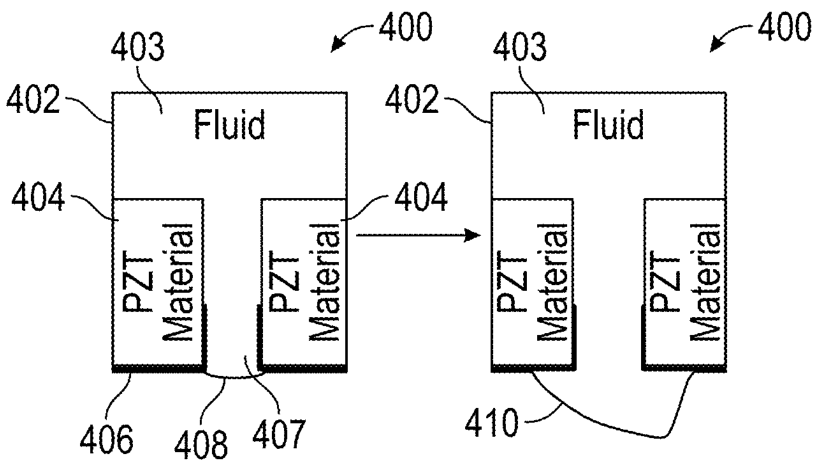

FIG. 7A

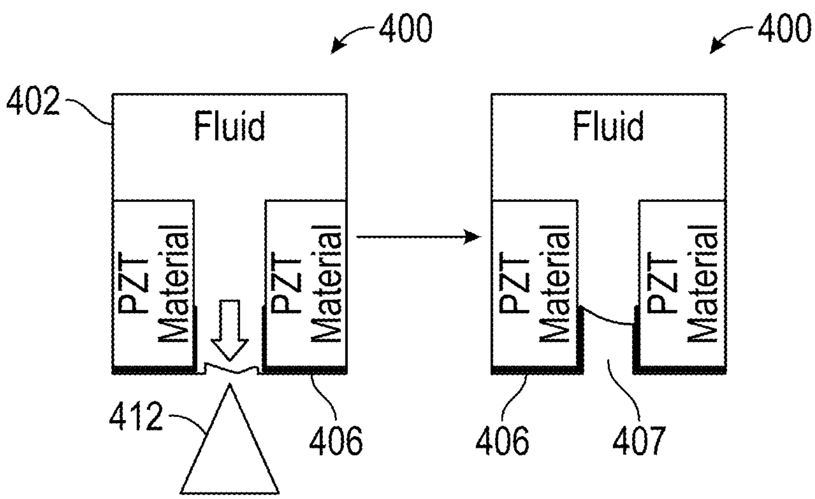

FIG. 7B

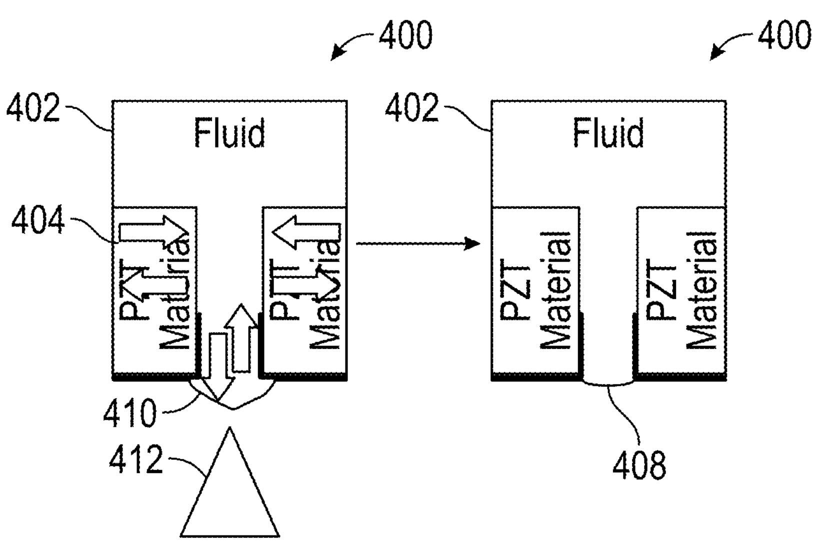

FIG. 7C

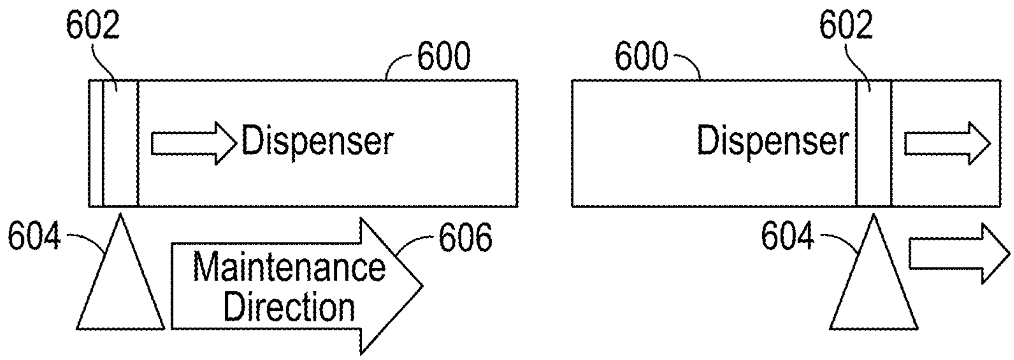
602
600
Dispenser
604
Maintenance
Direction
606
600
602
Dispenser
604

FIG. 8

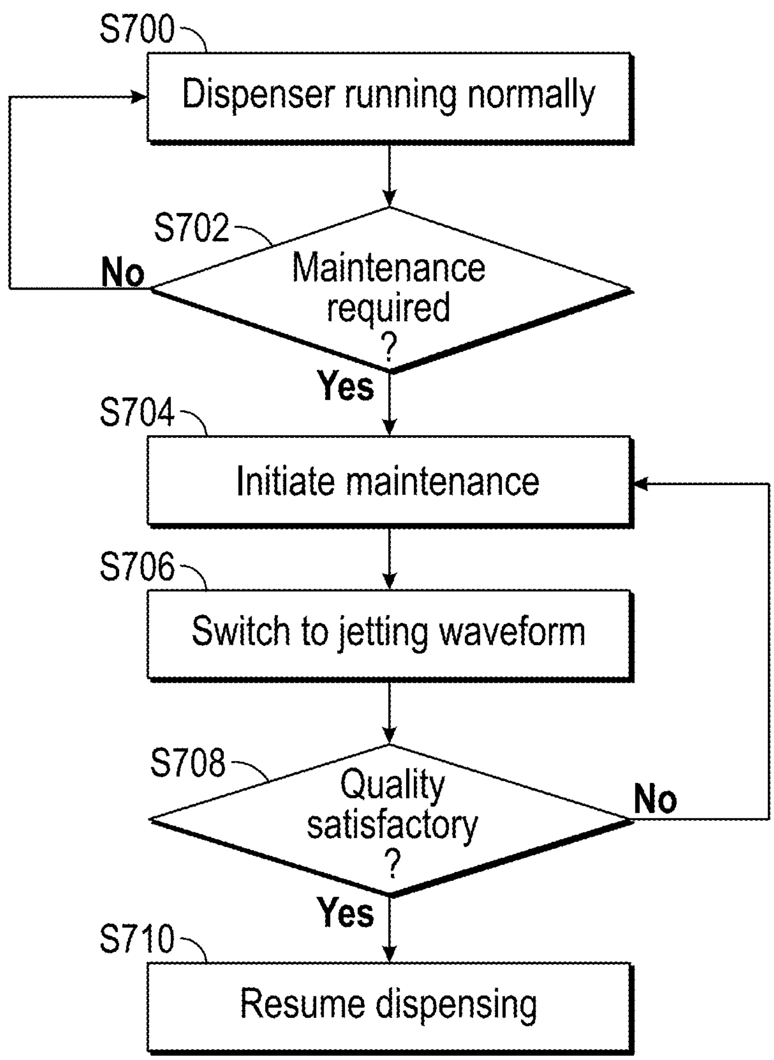
S700
Dispenser running normally
S702
Maintenance
required
?
No
Yes
S704
Initiate maintenance
S706
Switch to jetting waveform
S708
Quality
satisfactory
?
No
Yes
S710
Resume dispensing

FIG. 9

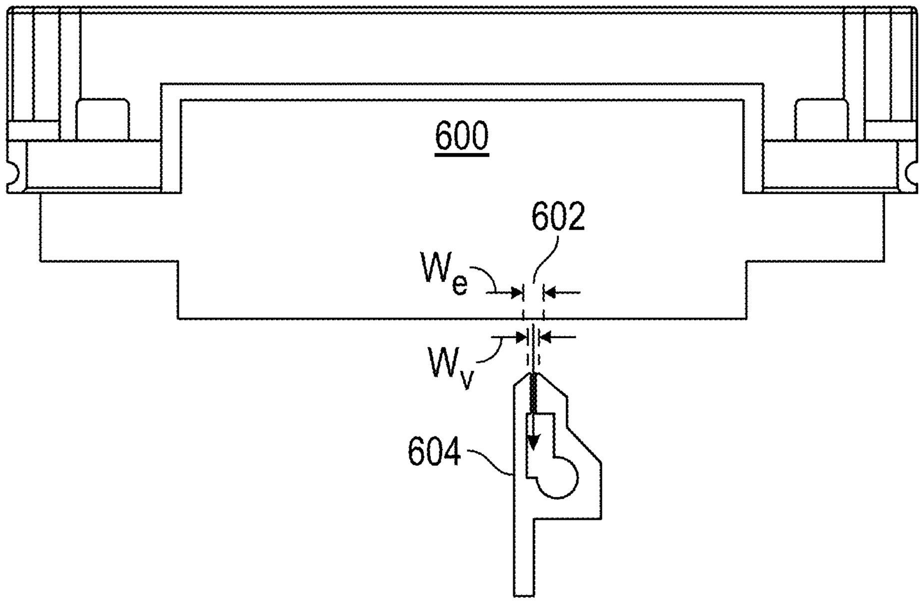

FIG. 10

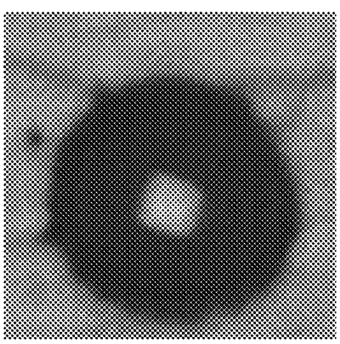

FIG. 11A

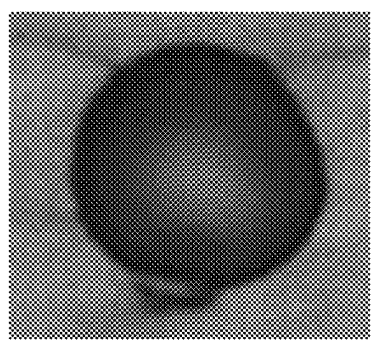

FIG. 11B

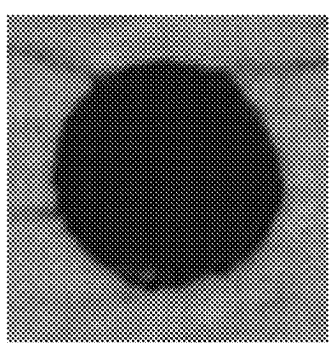

FIG. 11C

METHOD OF CLEANING FLUID DISPENSER BY APPLYING SUCTION FORCE AND VIBRATING MENISCUS

BACKGROUND

Field of Art

The present disclosure relates to a method of cleaning a dispenser during non-contact maintenance of the dispenser, in particular, the cleaning method may be applied to a dispenser associated with an ink jet head.

Description of the Related Art

Nano-fabrication includes the fabrication of very small structures that have features on the order of 100 nanometers or smaller. One application in which nano-fabrication has had a sizeable impact is in the fabrication of integrated circuits. The semiconductor processing industry continues to strive for larger production yields while increasing the circuits per unit area formed on a substrate. Improvements in nano-fabrication include providing greater process control and/or improving throughput while also allowing continued reduction of the minimum feature dimensions of the structures formed.

One nano-fabrication technique in use today is commonly referred to as nanoimprint lithography. Nanoimprint lithography is useful in a variety of applications including, for example, fabricating one or more layers of integrated devices by shaping a film on a substrate. Examples of an integrated device include but are not limited to CMOS logic, microprocessors, NAND Flash memory, NOR Flash memory, DRAM memory, MRAM, 3D cross-point memory, Re-RAM, Fe-RAM, STT-RAM, MEMS, and the like. Exemplary nanoimprint lithography systems and processes are described in detail in numerous publications, such as U.S. Pat. Nos. 8,349,241, 8,066,930, and 6,936,194, all of which are hereby incorporated by reference herein.

The nanoimprint lithography technique disclosed in each of the aforementioned patents describes the shaping of a film on a substrate by the formation of a relief pattern in a formable material (polymerizable) layer. The shape of this film may then be used to transfer a pattern corresponding to the relief pattern into and/or onto an underlying substrate.

The patterning process uses a template spaced apart from the substrate and the formable material is applied between the template and the substrate. The template is brought into contact with the formable material causing the formable material to spread and fill the space between the template and the substrate. The formable liquid is solidified to form a film that has a shape (pattern) conforming to a shape of the surface of the template that is in contact with the formable liquid. After solidification, the template is separated from the solidified layer such that the template and the substrate are spaced apart.

The substrate and the solidified layer may then be subjected to additional processes, such as etching processes, to transfer an image into the substrate that corresponds to the pattern in one or both of the solidified layer and/or patterned layers that are underneath the solidified layer. The patterned substrate can be further subjected to known steps and processes for device (article) fabrication, including, for example, curing, oxidation, layer formation, deposition, doping, planarization, etching, formable material removal, dicing, bonding, and packaging, and the like.

The nano-fabrication technique involves dispensing the formable material from a dispenser onto the substrate. Over many dispensing cycles, the formable material may begin to accumulate on a faceplate of the dispenser. Eventually, the amount of accumulation can interfere with the production and requires maintenance. A cleaning method and cleaning system that does not physically contact the faceplate when cleaning the formable material accumulated on the faceplate of the dispenser is typically preferred. However, a cleaning method and cleaning system that does not physically contact the faceplate uses a suction apparatus. If the suction apparatus is too close to the faceplate or the suction force/power is too strong, it may cause a meniscus associated with a nozzle of the faceplate to break. The meniscus is a curved upper surface of the fluid in the nozzle. Thus, there is a need in the art for a cleaning method and system that does not physically contact the faceplate while preventing breakage of the meniscus associated with nozzles of the faceplate of the fluid dispenser.

SUMMARY

The present disclosure includes a method for cleaning a fluid dispenser during non-contact maintenance of the fluid dispenser without breaking the meniscus associated with each nozzle of the fluid dispenser.

A method of cleaning a fluid dispenser for dispensing a material during non-contact maintenance of the fluid dispenser. The fluid dispenser including a plurality of nozzles disposed on a faceplate. The method including applying a suction force onto a surface of the faceplate using a suction apparatus, the suction apparatus being translated from one end of the faceplate to an opposite end of the faceplate such that a portion of nozzles from the plurality of nozzles are exposed to the suction force. The method continues by vibrating a menisci of the portion of nozzles that are exposed to the suction force to remove at least a portion of the material accumulated on the faceplate.

A dispensing system including a fluid dispenser configured to dispense a material, with a faceplate, a plurality of nozzles and a suction apparatus for applying a suction force onto a faceplate. The dispensing system also including one or more processors and one or more memories storing instructions, when executed by the one or more processors, causes the dispensing system to apply the suction force onto a surface of the faceplate using the suction apparatus, the suction apparatus being translated from one end of the faceplate to an opposite end of the faceplate such that a portion of nozzles from the plurality of nozzles are exposed to the suction force and vibrating a menisci of the portion of nozzles that are exposed to the suction force to remove at least a portion of the material accumulated on the faceplate.

A method of making an article including cleaning a fluid dispenser with a faceplate having a plurality of nozzles, the cleaning including applying a suction force onto a surface of the faceplate using a suction apparatus, the suction apparatus being translated from one end of the faceplate to an opposite end of the faceplate such that a portion of nozzles from the plurality of nozzles are exposed to the suction force, and vibrating a menisci of the portion of nozzles that are exposed to the suction force to remove at least a portion of material accumulated on the faceplate. The method of making an article continues by dispensing a portion of the material onto a substrate using the fluid dispenser, forming a pattern or a layer of the dispensed material on the substrate and processing the formed pattern or layer to make the article.

These and other objects, features, and advantages of the present disclosure will become apparent upon reading the following detailed description of exemplary embodiments of the present disclosure, when taken in conjunction with the appended drawings, and provided claims.

BRIEF DESCRIPTION OF DRAWINGS

So that features and advantages of the present disclosure can be understood in detail, a more particular description of embodiments of the disclosure may be had by reference to the embodiments illustrated in the appended drawings. It is to be noted, however, that the appended drawings only illustrate typical embodiments of the disclosure, and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

FIG. **1** is an illustration of an exemplary nanoimprint lithography system in accordance with an example embodiment.

FIG. **2** is a flowchart illustrating an exemplary imprinting method in accordance with an example embodiment.

FIG. **3**A shows a side view of a dispenser in accordance with an example embodiment.

FIG. **3**B shows an underside view of the dispenser of FIG. **3**A in accordance with an example embodiment.

FIG. **4**A shows a side view of the dispenser after formable material has accumulated on the surface of a faceplate, in accordance with an example embodiment.

FIG. **4**B shows a bottom view of the dispenser of FIG. **4A** after formable material has accumulated on the surface of the faceplate, in accordance with an example embodiment.

FIG. **5** shows a perspective exploded view of a vacuum apparatus in accordance with an example embodiment.

FIG. **6** shows a schematic of nozzles performing as expected before non-contact maintenance and several nozzle outages after non-contact maintenance.

FIG. **7**A shows a schematic of a nozzle including an ink chamber in good condition and a nozzle that requires maintenance.

FIG. **7**B shows a schematic of the nozzle from FIG. **7**A that requires maintenance where non-contact maintenance using a vacuum apparatus causing a broken meniscus in the nozzle.

FIG. **7**C shows a schematic of the nozzle from FIG. **7**A that requires maintenance where non-contact maintenance using a vacuum apparatus, in accordance with an example embodiment.

FIG. **8** shows a schematic emphasizing a selective area with meniscus excitation during maintenance, in accordance with an example embodiment.

FIG. **9** illustrates a block diagram showing white pixel jetting during non-contact maintenance of the dispenser in accordance with an example embodiment.

FIG. **10** shows a schematic emphasizing a width of a selective area with meniscus excitation during maintenance relative to a characteristic width of a vacuum apparatus, in accordance with an example embodiment.

FIGS. **11**A-C are micrographs menisci in various states.

Throughout the figures, the same reference numerals and characters, unless otherwise stated, are used to denote like features, elements, components or portions of the illustrated embodiments. Moreover, while the subject disclosure will now be described in detail with reference to the figures, it is done so in connection with the illustrative exemplary embodiments. It is intended that changes and modifications can be made to the described exemplary embodiments without departing from the true scope and spirit of the subject disclosure as defined by the appended claims.

DETAILED DESCRIPTION

Throughout this disclosure, reference is made primarily to nanoimprint lithography, which uses the above-mentioned patterned template to impart a pattern onto formable liquid. However, as mentioned below, in an alternative embodiment, the template is featureless in which case a planar surface may be formed on the substrate. In such embodiments where a planar surface is formed, the formation process is referred to as planarization. Thus, throughout this disclosure, whenever nanoimprint lithography is mentioned, it should be understood that the same method is applicable to planarization. The term superstrate is used in place of the term template in instances where the template is featureless.

Advances in inkjet technology has facilitated fabrication processes across a wide range of technology. Nanoimprint lithography is one technology that uses inkjet heads for its process. Inkjet heads eject drops of fluid (for example photoresist) from microscopic nozzles and deposit them on substrates. The drops targeted to a surface to fabricate three-dimensional (3D) structures such as electronic components as well as materials for applications in life sciences. Inkjet technology is popular in the development of nanotechnology because it can precisely deposit picoliter volumes of solutions or suspensions in well-defined patterns. The fluid deposited is a functional material such as a photoresist. One method is a single drop on demand (DOD). An electric voltage signal is applied to a piezoelectric (PZT) transducer to cause mechanical deformations of a fluid chamber to squeeze the fluid out of the nozzle and form a drop, mimicking the manner in which gravity causes the liquid to drip from larger nozzles. Volume and velocity of individual drops and the time interval between two successive drops can be fully controlled by the appropriate adjustment of voltage signals, such as waveforms, voltage amplitudes and voltage durations.

A nano-fabrication technique using inkjet heads involves dispensing formable material from a dispenser onto a substrate. However, over many dispensing cycles, the formable material may accumulate on a faceplate of the dispenser. Eventually, the amount of accumulation can interfere with the production and needs cleaning. The present disclosure is concerned with a non-contact air cleaning method that avoids breaking the meniscus of nozzles located on the faceplate of the dispenser. The fluid or formable material dispenser exists within a nanoimprint lithography system discussed below with reference to FIG. **1**.

FIG. **1** is an illustration of a nanoimprint lithography system **100**. The nanoimprint lithography system **100** is used to shape a film on a substrate **102**. The substrate **102** may be coupled to a substrate chuck **104**. The substrate chuck **104** may be but is not limited to a vacuum chuck, pin-type chuck, groove-type chuck, electrostatic chuck, electromagnetic chuck, and/or the like.

The substrate **102** and the substrate chuck **104** may be further supported by a substrate positioning stage **106**. The substrate positioning stage **106** may provide translational and/or rotational motion along one or more of the x, y, z, θ, and φ-axes. The substrate positioning stage **106**, the substrate **102**, and the substrate chuck **104** may also be positioned on a base (not shown). The substrate positioning stage may be a part of a positioning system.

Spaced-apart from the substrate **102** is a template **108**. The template **108** may include a body having a mold **110** extending towards the substrate **102** on a front side of the template **108**. The mold **110** may have a patterning surface **112** thereon also on the front side of the template **108**. Alternatively, the template **108** may be formed without the mold **110**, in which case the surface of the template facing the substrate **102** is equivalent to the mold **110** and the patterning surface **112** is that surface of the template **108** facing the substrate **102**.

The template **108** may be formed from such materials including, but not limited to, fused-silica, quartz, silicon, organic polymers, siloxane polymers, borosilicate glass, fluorocarbon polymers, metal, hardened sapphire, and/or the like. The patterning surface **112** may have features defined by a plurality of spaced-apart template recesses **114** and/or template protrusions **116**. The patterning surface **112** defines a pattern that forms the basis of a pattern to form on the substrate **102**. In an alternative embodiment, the patterning surface **112** is featureless in which case a planar surface is formed on the substrate. In an alternative embodiment, the patterning surface **112** is featureless and the same size as the substrate and a planar surface is formed across the entire substrate. In such embodiments where a planar surface is formed, the formation process may be alternatively referred to as planarization and the featureless template may be alternatively referred to as a superstrate.

Template **108** may be coupled to a template chuck **118**. The template chuck **118** may be, but is not limited to, vacuum chuck, pin-type chuck, groove-type chuck, electrostatic chuck, electromagnetic chuck, and/or other similar chuck types. The template chuck **118** may be configured to apply stress, pressure, and/or strain to template **108** that varies across the template **108**. The template chuck **118** may include piezoelectric actuators which can squeeze and/or stretch different portions of the template **108**. The template chuck **118** may include a system such as a zone based vacuum chuck, an actuator array, a pressure bladder, etc. which can apply a pressure differential to a back surface of the template causing the template to bend and deform.

The template chuck **118** may be coupled to an imprint head **120** which is a part of the positioning system. The imprint head **120** may be moveably coupled to a bridge. The imprint head may include one or more actuators such as voice coil motors, piezoelectric motors, linear motor, nut and screw motor, etc, which are configured to move the template chuck **118** relative to the substrate in at least the z-axis direction, and potentially other directions (e.g. x, y, θ, ψ, and (φ-axes).

The nanoimprint lithography system **100** further comprises a fluid dispenser **122**. The fluid dispenser **122** may also be moveably coupled to the bridge. In an embodiment, the fluid dispenser **122** and the imprint head **120** share one or more or all positioning components. In an alternative embodiment, the fluid dispenser **122** and the imprint head **120** move independently from each other. The fluid dispenser **122** may be used to deposit liquid formable material **124** (e.g., polymerizable material) onto the substrate **102** in a pattern. Additional formable material **124** may also be added to the substrate **102** using techniques, such as, drop dispense, spin-coating, dip coating, chemical vapor deposition (CVD), physical vapor deposition (PVD), thin film deposition, thick film deposition, and/or the like prior to the formable material **124** being deposited onto the substrate **102**. The formable material **124** may be dispensed upon the substrate **102** before and/or after a desired volume is defined between the mold **110** and the substrate **102** depending on design considerations. The formable material **124** may comprise a mixture including a monomer as described in U.S. Pat. Nos. 7,157,036 and 8,076,386, both of which are herein incorporated by reference.

Different fluid dispensers **122** may use different technologies to dispense formable material **124**. When the formable material **124** is jettable, ink jet type dispensers dispense the formable material **124**. For example, thermal ink jetting, microelectromechanical systems (MEMS) based ink jetting, valve jet, and piezoelectric ink jetting are common techniques for dispensing jettable liquids.

The nanoimprint lithography system **100** may further comprise a radiation source **126** that directs actinic energy along an exposure path **128**. The imprint head **120** and the substrate positioning stage **106** may be configured to position the template **108** and the substrate **102** in superimposition with the exposure path **128**. The radiation source **126** sends the actinic energy along the exposure path **128** after the template **108** has made contact with the formable material **128**. FIG. **1** illustrates the exposure path **128** when the template **108** is not in contact with the formable material **124**, this is done for illustrative purposes so that the relative position of the individual components can be easily identified. An individual skilled in the art would understand that exposure path **128** would not substantially change when the template **108** is brought into contact with the formable material **124**.

The nanoimprint lithography system **100** may further comprise a field camera **136** that is positioned to view the spread of formable material **124** after the template **108** has made contact with the formable material **124**. FIG. **1** illustrates an optical axis of the field camera's imaging field as a dashed line. As illustrated in FIG. **1** the nanoimprint lithography system **100** may include one or more optical components (dichroic mirrors, beam combiners, prisms, lenses, mirrors, etc.) which combine the actinic radiation with light to be detected by the field camera. The field camera **136** may be configured to detect the spread of formable material under the template **108**. The optical axis of the field camera **136** as illustrated in FIG. **1** is straight but may be bent by one or more optical components. The field camera **136** may include one or more of a CCD, a sensor array, a line camera, and a photodetector which are configured to gather light that has a wavelength that shows a contrast between regions underneath the template **108** that are in contact with the formable material, and regions underneath the template **108** which are not in contact with the formable material **124**. The field camera **136** may be configured to gather monochromatic images of visible light. The field camera **136** may be configured to provide images of the spread of formable material **124** underneath the template **108**, the separation of the template **108** from cured formable material, and can be used to keep track of the progress over the imprinting process.

The nanoimprint lithography system **100** may further comprise a droplet inspection system **138** that is separate from the field camera **136**. The droplet inspection system **138** may include one or more of a CCD, a camera, a line camera, and a photodetector. The droplet inspection system **138** may include one or more optical components such as lenses, mirrors, apertures, filters, prisms, polarizers, windows, adaptive optics, and/or light sources. The droplet inspection system **138** may be positioned to inspect droplets prior to the patterning surface **112** contacting the formable material **124** on the substrate **102**.

The nanoimprint lithography system **100** may further include a thermal radiation source **134** which may be configured to provide a spatial distribution of thermal radiation to one or both of the template **108** and the substrate **102**. The thermal radiation source **134** may include one or more sources of thermal electromagnetic radiation that will heat up one or both of the substrate **102** and the template **108** and does not cause the formable material **124** to solidify. The thermal radiation source **134** may include a spatial light modulator such as a digital micromirror device (DMD), Liquid Crystal on Silicon (LCoS), Liquid Crystal Device (LCD), etc., to modulate the spatial temporal distribution of thermal radiation. The nanoimprint lithography system **100** may further comprise one or more optical components which are used to combine the actinic radiation, the thermal radiation, and the radiation gathered by the field camera **136** onto a single optical path that intersects with the imprint field when the template **108** comes into contact with the formable material **124** on the substrate **102**. The thermal radiation source **134** may send the thermal radiation along a thermal radiation path (which in FIG. **1** is illustrated as 2 thick dark lines) after the template **108** has made contact with the formable material **124**. FIG. **1** illustrates the thermal radiation path when the template **108** is not in contact with the formable material **124**, this is done for illustrative purposes so that the relative position of the individual components can be easily identified. An individual skilled in the art would understand that the thermal radiation path would not substantially change when the template **108** is brought into contact with the formable material **124**. In FIG. **1** the thermal radiation path is shown terminating at the template **108**, but it may also terminate at the substrate **102**. In an alternative embodiment, the thermal radiation source **134** is underneath the substrate **102**, and thermal radiation path is not combined with the actinic radiation and visible light.

Prior to the formable material **124** being dispensed onto the substrate **102**, a substrate coating **132** may be applied to the substrate **102**. In an embodiment, the substrate coating **132** may be an adhesion layer. In an embodiment, the substrate coating **132** may be applied to the substrate **102** prior to the substrate being loaded onto the substrate chuck **104**. In an alternative embodiment, the substrate coating **132** may be applied to substrate **102** while the substrate **102** is on the substrate chuck **104**. In an embodiment, the substrate coating **132** may be applied by spin coating, dip coating, etc. In an embodiment, the substrate **102** may be a semiconductor wafer. In another embodiment, the substrate **102** may be a blank template (replica blank) that may be used to create a daughter template after being imprinted.

The nanoimprint lithography system **100** may be regulated, controlled, and/or directed by one or more processors **140** (controller) in communication with one or more components and/or subsystems such as the substrate chuck **104**, the substrate positioning stage **106**, the template chuck **118**, the imprint head **120**, the fluid dispenser **122**, the radiation source **126**, the thermal radiation source **134**, the field camera **136** and/or the droplet inspection system **138**. The processor **140** may operate based on instructions in a computer readable program stored in a non-transitory computer readable memory **142**. The processor **140** may be or include one or more of a CPU, MPU, GPU, ASIC, FPGA, DSP, and a general purpose computer. The processor **140** may be a purpose built controller or may be a general purpose computing device that is adapted to be a controller. Examples of a non-transitory computer readable memory include but are not limited to RAM, ROM, CD, DVD, Blu-Ray, hard drive, networked attached storage (NAS), an intranet connected non-transitory computer readable storage device, and an internet connected non-transitory computer readable storage device.

Either the imprint head **120**, the substrate positioning stage **106**, or both varies a distance between the mold **110** and the substrate **102** to define a desired space (a bounded physical extent in three dimensions) that is filled with the formable material **124**. For example, the imprint head **120** may apply a force to the template **108** such that mold **110** is in contact with the formable material **124**. After the desired volume is filled with the formable material **124**, the radiation source **126** produces actinic radiation (e.g. UV, 248 nm, 280 nm, 350 nm, 365 nm, 395 nm, 400 nm, 405 nm, 435 nm, etc.) causing formable material **124** to cure, solidify, and/or cross-link; conforming to a shape of the substrate surface **130** and the patterning surface **112**, defining a patterned layer on the substrate **102**. The formable material **124** is cured while the template **108** is in contact with formable material **124** forming the patterned layer on the substrate **102**. Thus, the nanoimprint lithography system **100** uses an imprinting process to form the patterned layer which has recesses and protrusions which are an inverse of the pattern in the patterning surface **112**. In an alternative embodiment, the nanoimprint lithography system **100** uses an imprinting process to form the planar layer with a featureless patterning surface **112**.

The imprinting process may be done repeatedly in a plurality of imprint fields that are spread across the substrate surface **130**. Each of the imprint fields may be the same size as the mold **110** or just the pattern area of the mold **110**. The pattern area of the mold **110** is a region of the patterning surface **112** which is used to imprint patterns on a substrate **102** which are features of the device or are then used in subsequent processes to form features of the device. The pattern area of the mold **110** may or may not include mass velocity variation features which are used to prevent extrusions. In an alternative embodiment, the substrate **102** has only one imprint field which is the same size as the substrate **102** or the area of the substrate **102** which is to be patterned with the mold **110**. In an alternative embodiment, the imprint fields overlap. Some of the imprint fields may be partial imprint fields which intersect with a boundary of the substrate **102**.

The patterned layer may be formed such that it has a residual layer having a residual layer thickness (RLT) that is a minimum thickness of formable material **124** between the substrate surface **130** and the patterning surface **112** in each imprint field. The patterned layer may also include one or more features such as protrusions which extend above the residual layer having a thickness. These protrusions match the recesses **114** in the mold **110**.

FIG. **2** is a flowchart of an imprinting process by the nanoimprint lithography system **100** that can be used to form patterns in formable material **124** on one or more imprint fields (also referred to as: pattern areas or shot areas). The imprinting process may be performed repeatedly on a plurality of substrates **102** by the nanoimprint lithography system **100**. The processor **140** may be used to control the imprinting process.

In an alternative embodiment, a similar process may be performed to planarize the substrate **102**. In the case of planarizing, substantially the same steps discussed herein with respect to FIG. **2** are performed, except that a patternless superstrate is used in place of the template. Thus, it should be understood that the following description is also applicable to a planarizing method. When using as superstrate, the superstrate may be the same size or larger than the substrate **102**.

The beginning of the imprinting process may include a template mounting step causing a template conveyance mechanism to mount a template **108** onto the template chuck **118**. The imprinting process may also include a substrate mounting step, the processor **140** may cause a substrate conveyance mechanism to mount the substrate **102** onto the substrate chuck **104**. The substrate may have one or more coatings and/or structures. The order in which the template **108** and the substrate **102** are mounted onto the nanoimprint lithography system **100** is not particularly limited, and the template **108** and the substrate **102** may be mounted sequentially or simultaneously.

In a positioning step, the processor **140** may cause one or both of the substrate positioning stage **106** and/or a dispenser positioning stage to move an imprint field i (index i may be initially set to 1) of the substrate **102** to a fluid dispense position below the fluid dispenser **122**. The substrate **102**, may be divided into N imprint fields, wherein each imprint field is identified by an index i. In which N is a real integer such as 1, 10, 75, etc. $\{N \in \mathbb{Z}^+\}$. In a dispensing step S**202**, the processor **140** may cause the fluid dispenser **122** to dispense formable material onto an imprint field i. In an embodiment, the fluid dispenser **122** dispenses the formable material **124** as a plurality of droplets. The fluid dispenser **122** may include one nozzle or multiple nozzles. The fluid dispenser **122** may eject formable material **124** from the one or more nozzles simultaneously. The imprint field i may be moved relative to the fluid dispenser **122** while the fluid dispenser is ejecting formable material **124**. Thus, the time at which some of the droplets land on the substrate may vary across the imprint field i. In an embodiment, during the dispensing step S**202**, the formable material **124** may be dispensed onto a substrate in accordance with a drop pattern. The drop pattern may include information such as one or more of position to deposit drops of formable material, the volume of the drops of formable material, type of formable material, shape parameters of the drops of formable material, etc.

After, the droplets are dispensed, then a contacting step S**204** may be initiated, the processor **140** may cause one or both of the substrate positioning stage **106** and a template positioning stage to bring the patterning surface **112** of the template **108** into contact with the formable material **124** in imprint field i.

During a spreading step S**206**, the formable material **124** then spreads out towards the edge of the imprint field i and the mold sidewalls. The edge of the imprint field may be defined by the mold sidewalls. How the formable material **124** spreads and fills the mold can be observed via the field camera **136** and may be used to track a progress of a fluid front of formable material.

In a curing step S**208**, the processor **140** may send instructions to the radiation source **126** to send a curing illumination pattern of actinic radiation through the template **108**, the mold **110** and the patterning surface **112**. The curing illumination pattern provides enough energy to cure (polymerize) the formable material **124** under the patterning surface **112**.

In a separation step S**210**, the processor **140** uses one or more of the substrate chuck **104**, the substrate positioning stage **106**, template chuck **118**, and the imprint head **120** to separate the patterning surface **112** of the template **108** from the cured formable material on the substrate **102**.

If there are additional imprint fields to be imprinted then the process moves back to step S**202**. In an embodiment, additional processing is performed on the substrate **102** in a processing step S**212** to create an article of manufacture (e.g. semiconductor device) by forming a pattern or a layer of the dispensed material on the substrate and processing the formed pattern or layer to make the article. In an embodiment, each imprint field includes a plurality of devices.

The further processing in processing step S**212** may include etching processes to transfer a relief image into the substrate that corresponds to the pattern in the patterned layer or an inverse of that pattern. The further processing in processing step S**212** may also include known steps and processes for article fabrication, including, for example, curing, oxidation, layer formation, deposition, doping, planarization, etching, formable material removal, dicing, bonding, and packaging, and the like. The substrate **102** may be processed to produce a plurality of articles (devices).

Referring now to FIG. **3**A, a side view of the dispenser **122** is shown. The dispenser **122** includes a faceplate **133** with a first end **123** and a second end **125**. The faceplate **133** has a length **127** extending in an X dimension from the first end **123** to the second end **125**. FIG. **3**B shows an underside view of the dispenser **122**. The underside view of the dispenser **122** reveals a surface **135** of the faceplate **133** in which a plurality of dispensing nozzles **137** are formed. The number of nozzles **137** formed on the surface **135** of the faceplate **133** may be on the order of hundreds, for example 500 or more. The faceplate **133** includes a width **139** extending in a Y dimension, the Y dimension being perpendicular to the X dimension.

FIGS. **4**A and **4**B show several views of the dispenser **122** after formable material **124** has accumulated on the surface **135** of the faceplate **133**. The structure of the dispenser **122** is the same as discussed above with respect to FIGS. **3**A and **3**B, the only difference being that formable material **124** has accumulated on the surface **135** of the faceplate **133** after the dispenser **122** has dispensed formable material **124** many times. As shown schematically in FIGS. **4**A and **4**B the accumulated formable material **124** may be located with various patterns and thicknesses across the surface **135** of the faceplate **133**. The formable material **124** on the surface **135** of the faceplate **133** may cause outages of nozzles when dispensing the formable material. In order to remove the resist or accumulated formable material **124** from the surface **135** of the faceplate **133** a cleaning process is applied to the dispenser **122**. The cleaning process may use a non-contact cleaning method for example a vacuum apparatus (suction apparatus) may be used to remove formable material **124** from the surface **135** of the faceplate **133**. The vacuum apparatus is described in further detail below with reference to FIG. **5**.

Referring now to FIG. **5** is an exploded view of an exemplary vacuum apparatus that may be used to impart a suction force on the surface **135** of the faceplate **133** of the dispenser **122**. The vacuum apparatus **204** is used to clean formable material from the dispenser **122**. The vacuum apparatus **204** is coupled with a vacuum source **205**. The vacuum apparatus **204** may be translated across the surface **135** of the faceplate **133** using a translating mechanism (not shown). The vacuum apparatus **204** may be translated along the X-axis as shown in FIGS. **3**A and **3**B. The reason for translating the vacuum apparatus is to ensure the formable material may be removed from the entire surface of the faceplate **133**.

The translating mechanism may be any mechanism known in the art that is suitable for imparting linear translation of an object. For example, the translating mechanism may be a linear actuator that may include: a stepper motor; a linear motor; a moving coil; a hydraulic actuator; pneumatic actuator; and the like. The linear actuator may include a position encoder. The position encoder may be a rotary or linear encoder. Such encoders are known in the art and provide position information at a particular moment in time.

The vacuum apparatus **204** includes a vacuum orifice **206**. The vacuum orifice **206** may be an elongated slit shape where the length of the vacuum orifice **206** is much longer than the width of the vacuum orifice **206**. For example, the ratio of the length of the vacuum orifice **206** to the width of the vacuum office **206** may be from 5:1 to 100:1, may be 10:1 to 90:1, may be 20:1 to 80:1, may be 30:1 to 70:1, or may be 40:1 to 65:1. In one example, the ratio may be 60:1. In an example embodiment, the length of the vacuum orifice **206** is substantially the same (e.g., within ±20%) as the width **139** of the faceplate **133** in the Y dimension. In an example embodiment, the width of the vacuum orifice **206** is wide enough so that it does not become clogged with formable material while being narrow enough to supply sufficient vacuum (for example, 0.5 mm, 1 mm, etc.). The vacuum orifice **206** may also have lips on each side which are similar (e.g., within ±30%) in width to the width of the vacuum orifice **206** which helps constrain the suction force to a local region of the faceplate **133**. The vacuum apparatus **204** may further include a vacuum connector **212** and a connector port **210**. The vacuum connector **212** may include a first end **214** that connects with the connector port **210** and a second end **216** that connects with the vacuum source **205**. Thus, by activating the vacuum source **205**, a suction force is applied to the vacuum orifice **206**.

A vacuum force can be selectively imparted on the faceplate **133** by actuating the vacuum source **205** when the vacuum orifice **206** of the vacuum apparatus **204** travels along the X dimension. The orifice **206** of the vacuum apparatus **204** also extends across the width **139** of the faceplate **133**.

Using the vacuum apparatus **204** for cleaning the formable material **124** from the surface **135** of the faceplate **133** is known as non-contact maintenance. When performing non-contact maintenance, occasionally nozzle outages are observed after maintenance is completed. This appears to occur when the suction force i.e. vacuum level is too high or the vacuum apparatus **204** is too close to the dispenser **122**. Also, it is prevalent and common when using dispensers with a modified surface **135** of the faceplate **133** in which the photoresist is designed to not wet the surface **135** after the surface **135** has been modified. A non-wetting surface has a relatively low surface energy which can induce formable material that has accumulated on the surface **135** into forming large beads on the surface **135**. Nozzle outages after non-contact maintenance may be recovered with a purge or fluid (change of meniscus pressure), it is assumed that the meniscus has been broken and needs to be reestablished. To mitigate this, the meniscus may be directly excited during jetting using a non-jetting waveform. A non-jetting waveform is also known as white pixel jetting. The non-jetting waveform is used so that no fluid is ejected into the process module, less fluid accumulates on the surface **135**, and all dispenser nozzles dispense after maintenance. The amount of white pixel jetting should be minimized as it can cause some accumulation of fluid onto the surface **135**, so only the area actively undergoing maintenance should be excited with a non-jetting waveform. The meniscus can also be excited during maintenance in regular conditions if the jetting performance after regular maintenance is not satisfactory. Directly exciting the meniscus during non-contact maintenance may reduce nozzle outages from the faceplate cleaning process.

The applicant has found that it is advantageous to perform non-contact maintenance periodically to remove excess fluid from the dispenser faceplate **133**. The applicant has also found that performance of the nanoimprint lithography system **100** is improved if every nozzle is required to fire subsequently after the non-contact maintenance was performed. The applicant has found that a surface energy mismatch between the formable material **124** and surface **135** that is non-wetting increases the likelihood of a nozzle performance degradation after non-contact maintenance in the manner of the prior art is performed. The applicant has also found that this likelihood can be reduced if non-contact maintenance is performed while also exciting the meniscus of the nozzles as they are being cleaned. The applicant has also found that if the vacuum orifice drifts too close to the faceplate, the likelihood of nozzle performance degradation increases, this likelihood can be mitigated by exciting the meniscus of the nozzles as they are being cleaned. This improves the process window under which non-contact maintenance is performed.

A drop dispensing method by the nanoimprint lithography system **100** or planarization system can be used to dispense a pattern of drops of formable material **124** onto the substrate **102**, which is then imprinted/planarized. Imprinting/planarizing may be done in a field by field basis or on a whole wafer basis. The drops of formable material **124** may also be deposited in a field by field basis or on a whole substrate basis. Even when the drops are deposited on a whole substrate basis generating the drop pattern is preferably done on a field by field basis.

Generating a drop pattern for a full field may include a processor **140** receiving a substrate pattern of a representative substrate **102**, and a template pattern of a representative template **108**.

The substrate pattern may include information about substrate topography of the representative substrate, a field of the representative substrate and/or a full field of the representative substrate. The substrate topography may be measured, generated based on previous fabrication steps and/or generated based on design data. In an alternative embodiment, the substrate pattern is featureless either because there were no previous fabrication steps or the substrate had previously been planarized to reduce topography. The substrate topography may include information about the shape of an edge such as a beveled edge or a rounded edge of the representative substrate. The substrate topography may include information about the shape and position of one or more flats or notches which identify the orientation of the substrate. The substrate topography may include information about a shape and position of a reference edge which surrounds the area of the substrate on which patterns are to be formed.

The template pattern may include information about the topography of the patterning surface **112** of the representative template. The topography of the patterning surface **112** may be measured and/or generated based on design data. In an alternative embodiment, the template pattern of the representative embodiment is featureless and may be used to planarize the substrate **102**. The patterning surface **112** may be the same size as: an individual full field; multiple fields; the entire substrate, or larger than the substrate.

Once the substrate pattern and the template pattern are received, a processor **140** may calculate a distribution of formable material **124** that will produce a film that fills the volume between the substrate and the patterning surface when the substrate and the patterning surface are separated by a gap during imprinting. The distribution of formable material on the substrate may take the form of: an areal density of formable material; positions of droplets of formable material; and/or volume of droplets of formable material. Calculating the distribution of formable material may take into account one or more of: material properties of the formable material; material properties of the patterning surface; material properties of the substrate surface; spatial variation in volume between the patterning surface and the substrate surface; fluid flow; evaporation; etc.

An initial position of the vacuum apparatus **204** is set relative to the surface **135** of the faceplate **133**, prior to any accumulation of formable material on the faceplate **133**. That is, the faceplate **133** does not have any formable material on the surface **135**. The position of the vacuum apparatus **204** is changed relative to the surface **135** of the faceplate **133** by changing the angle of the vacuum apparatus **204** relative to the surface **135** of the faceplate **133**, the distance between the vacuum apparatus **204** and the faceplate **133**, or the angle of direction of travel of the vacuum apparatus **204** relative to the surface **135** of the faceplate **133**.

The vacuum apparatus **204** is used to clean the surface **135** of the faceplate **133**. The vacuum apparatus **204** may be mounted to a tray such that when a translation mechanism actuates, the tray, and the vacuum apparatus **204** all translate together across the surface **135** of the faceplate **133** in the X dimension. For this reason, it is possible to actuate the vacuum to suction formable material **124** off the surface **135** of the faceplate **133**. Accordingly, the vacuum apparatus **204** travels along the X dimension at a distance from the surface **135** of the faceplate **133** while sucking formable material **124** into the orifice **206**.

As the vacuum apparatus **204** travels along the X dimension from a first end **123** to a second end **125**, the vacuum pressure may be increased (become more negative) as the vacuum apparatus **204** approaches the second end **125**. The increase in the vacuum pressure may be from 25% higher to 100% higher, from 33% higher to 80% higher, or from 50% to 66% higher. The increase in vacuum pressure may be then held at the increased amount until reaching the second end **125** of the faceplate **133**. The benefit of increases in the vacuum pressure toward the second end **125** of the faceplate **133** is that the sudden increase assists in suctioning away formable material **124** that has been displaced during the vacuuming that has occurred up until this point in the X dimension. That is, during a time where the initial vacuum pressure is set, as the vacuum apparatus **204** travels across the faceplate **133** in the X dimension, some of the formable material **124** will be sucked up, while some will be displaced in a direction toward to the second end **125** of the faceplate **133**. While some of the displaced formable material **124** may be suctioned as the vacuum apparatus **204** continues to travel in the X dimension, other amounts will continue to displace in the direction of the second end **125** of the faceplate **133**. The sudden increase in vacuum pressure near the second end **125** of the faceplate **133** assists in suctioning off the final amount of displaced formable material **124**.

As described above, the nanoimprint lithography system **100** may be regulated, controlled, and/or directed by the one or more processors **140** (controller). This includes all of the method steps described above, including controlling the hardware that changes all three position factors that impact the position of the vacuum apparatus **204** relative to the faceplate **133**, controlling a translation mechanism to control the movement of the vacuum apparatus **204**, controlling when the vacuum pressure is applied and at what pressure. While not shown in the figures, it should be understood that any of the mechanical adjustments (i.e., adjusting the angle of approach, the angle of the vacuum apparatus relative to the faceplate, and the distance between the vacuum apparatus and the faceplate) can be controlled by the controller via a motor or other known automation means.

Referring now to FIG. **6**, a test pattern of dispensed drops is shown before and after non-contact maintenance of the dispenser **122**. The first test pattern of dispensed drops **300** shows an exemplary pattern of dispensed drops when all nozzles are firing as expected before maintenance of the dispenser **122**. The test pattern of dispensed drops is a pattern of drops dispensed onto a test substrate by the nanoimprint lithography tool **100** and is used to measure the performance of the fluid dispenser **122**. An expanded portion **310** of the test pattern of dispensed drops **300** is shown that indicates the droplets from the nozzles are not missing or deviated. The second test pattern of dispensed drops **320** is an example of a pattern of dispensed drops after non-contact maintenance is performed without white pixel jetting being performed at the same time. As can be observed in the expanded portion **330** of the test pattern of dispensed drops **320**, there are several nozzle outages (drops are missing or drops are at deviated positions) immediately after maintenance. Therefore, it is an important goal to use non-contact maintenance while avoiding nozzle outages such as missing fluid or deviated droplets from the nozzles.

Referring now to FIG. **7**A, a nozzle **400** in good condition on the left side and the same nozzle **400** in a condition where maintenance is required after regular jetting on the right side are shown. The nozzle **400** includes a fluid chamber **402** with fluid **403** or formable material disposed within the dispenser that is to be dispensed out of the nozzle **400**. In the embodiment shown in the present disclosure, the nozzle **400** includes a piezoelectric transducer (PZT) material **404**. Although PZT material is shown in FIG. **7**A, any well-known material may be used for the nozzle **400**. The PZT material **404** may include a non-wetting coating material **406** to prevent a surface portion of the PZT material **404** from accumulating fluid **403** on the surface **135** of the faceplate **133**. In an embodiment, the non-wetting coating material **406** has a higher surface energy than the surface **135** of the faceplate **133** that does not have the non-wetting coating material **406**. In an embodiment, the surface **135** is modified to be non-wetting for the formable material **403**. In an embodiment, a portion of the surface **135**, and/or the interior of the nozzle is non-wetting for the formable material **403**. In an embodiment, the non-wetting coating is a polyamide material. The surface of the fluid **403** within the nozzle **400** forms a meniscus **408**. The meniscus **408** may be relatively flush with an aperture **407** of the nozzle that is in good condition for jetting drops on a regular basis. In an embodiment, the meniscus is concave or convex and is located within the nozzle or is relatively flush with the aperture **407**. The position and shape of the meniscus **408** can be controlled by adjusting pressures of a supply pump and a return pump which control passage of fluid through the dispenser. The control over the position and shape of the meniscus for individual nozzles can be disrupted when an individual meniscus is broken. FIG. **11**A is an illustration of a micrograph of a convex meniscus. FIG. **11**B is a micrograph of a concave meniscus. FIG. **11**C is a micrograph of a broken meniscus.

The nozzle **400** on the right side requires maintenance because the accumulated formable material **410** has accumulated to a point that the likelihood of formable material dripping onto the substrate has become unacceptable. Typically, a nozzle requires maintenance after several hours, days or weeks of regular jetting. Regular jetting includes applying a voltage signal to the PZT material **404**. The voltage signal or jetting waveform causes mechanical deformations within the PZT material **404** which in turn cause the fluid/formable material **403** to be dispensed from the aperture **407** of the nozzle. Due to jetting which causes the fluid **403** to be dispensed from the nozzle, fluid **403** may accumulate at the surface of the nozzle after millions of drop dispensing cycles. When there is an accumulation of fluid or formable material **403** the nozzle requires maintenance to prevent deterioration during fluid dispensation. Thus, a vacuum apparatus is used for imparting a suction force on the nozzle to remove the fluid accumulation caused from jetting. In an embodiment, the walls of the nozzles may be made of PZT material. In an embodiment, one or more walls of the fluid chamber **402** that are connected to the nozzle **400** may be made of a PZT material. In an embodiment, the PZT transducer may cause mechanical deformations in a diaphragm connected to the fluid chamber or nozzle. In an embodiment, instead of using a PZT material to initiate jetting, a heater is used to generate bubbles to initiate jetting. In an embodiment, instead of using a PZT material to initiate jetting, a MEMS or other deformable device is used to initiate jetting.

Referring now to FIG. **7**B, the nozzle **400** on the left side has non-contact maintenance applied by the vacuum apparatus **412** that imparts a suction force on the accumulated formable material **410**. If the suction force is too great or if the vacuum apparatus **412** is too close to the nozzle **400**, then the non-contact maintenance may cause the meniscus **408** to break as shown in the nozzle **400** on the right side of FIG. **7**B. In an embodiment, the surface **135** of the faceplate is bowed or otherwise not flat, which can make it difficult to ensure that the vacuum apparatus **412** is the proper distance from the nozzle **400** as the nozzle travels across the faceplate. If the meniscus **408** is broken as shown, this may result in nozzle outages, drop volume variation, misshapen drops, and/or misalignment of drop landing positions. The present disclosure focuses on avoiding a broken meniscus as a result of non-contact maintenance.

Referring now to FIG. **7**C, non-contact maintenance is performed on the nozzle **400**, yet the meniscus **408** remains intact. FIG. **7**C shows that during non-contact maintenance of the nozzle **400** when the vacuum apparatus **412** is applied to remove the accumulated formable material **410**, meniscus excitation occurs by applying a voltage signal to the PZT material **404** causing the PZT material **404** to deform. The deformation of the PZT material **404** is such that the PZT material **404** expands and applies a force to the fluid **403**. The deformation of the PZT material **404** is enough to cause the PZT material **404** to expand perpendicular to a direction in which the fluid **403** is dispensed from the nozzle **400**. The force applied by the PZT material **404** should not exceed a threshold sufficient to cause fluid **403** to be dispensed from the nozzle **400**. The force applied by the PZT material **404** should be below the threshold that causes fluid **403** to be dispensed from the nozzle **400**. Applying a voltage to the PZT material **404** during non-contact maintenance of the nozzle **400** using a vacuum apparatus **412** may result in maintaining the meniscus **408** of the nozzle **400** while removing the normal accumulation **410** caused by jetting of the nozzles. In an alternative embodiment, the menisci of one or more nozzles is excited in a non-jetting manner by one or more of: PZT material that supplies a force to a fluid in a fluid nozzle; PZT material that supplies a force to a fluid in a chamber that is in fluid connection with a fluid nozzle; a MEMS device that supplies a force to fluid in fluid connection with a fluid nozzle; and a heater that creates a bubble that is in fluid communication with a fluid nozzle. The excitation of the meniscus or menisci may include applying a voltage to a jetting means corresponding to the portion of dispenser nozzle(s) such that the minimum amount of voltage required to vibrate the menisci without causing material to be dispensed from the portion of nozzles.

FIG. **8** is a schematic that emphasizes only a portion **602** of the dispenser **600** being excited during the non-contact maintenance. As each nozzle area is selectively excited as the vacuum apparatus **604** moves over a specific area of the dispenser. This allows for minimal meniscus excitation. Too much meniscus excitation may add to the accumulation of fluid to the faceplate. FIG. **8** shows the vacuum apparatus **604** moving in a direction **606** from the left side of the dispenser **600** to the right side of the dispenser **600** known as the maintenance direction **606**. The meniscus excitation is applied to an area on the dispenser **600** that is adjacent the vacuum apparatus **604** such that only a small portion of the dispenser **600** has the associated nozzles have a force applied by the PZT material to cause meniscus excitation. For example, the vacuum apparatus **604** may have a rectangular vacuum orifice in which the narrowest width is 0.5 mm. The vacuum apparatus **604** may also have 0.3 mm lips which surround the rectangular orifice. The vacuum apparatus may have a characteristic width (w_v) which includes the width of the narrowest width of the vacuum orifice and twice the width of the lips which in this example is 1.1 mm. Only a small portion of the dispenser **600** within an excited width (w_e) has the associated nozzles have their menisci excited. In an embodiment, the excited width is greater than the characteristic width ($w_e > w_v$) as illustrated in FIG. **10**. The excited width **602** is also limited by the pitch of the nozzle dispensers such that less than 2, 3, 4, 5, 6, 7, 8, 9 or 10 nozzles are excited at any one time.

FIG. **9** is a flow chart illustrating exemplary steps before, during and after non-contact maintenance to show when white pixel jetting is implemented to prevent meniscus breakage in the fluid dispensing nozzles in accordance with an embodiment of the present disclosure. In the first block **S700**, the dispenser is running normally which may cause normal accumulation on the surface of the nozzle. In the second block **S702** it is determined whether the dispenser requires maintenance from normal formable material accumulation. If the dispenser does not need to be maintained (insufficient accumulation), the following step may return to the first block **S700** and the dispenser may continue to run normally (jetting). Alternatively, if there is sufficient accumulation and it is determined that the dispenser requires maintenance, the next step is to retract dispenser and begin the maintenance process **S704**. The white pixel waveform is loaded in and maintenance is initiated.

During the maintenance step **S704**, the vacuum apparatus imparts a suction force on a partial area of the dispenser. The partial area of the dispenser exposed to the suction force is the area wherein the nozzles are excited to avoid breakage of the meniscus. After completing the maintenance of the dispenser, in step **S706** the white pixel waveform is switched to a jetting waveform to dispense fluid from the nozzles as a test for quality. If the dispenser quality is satisfactory, yes in step **S708**, the dispenser may proceed with dispensing the formable material in step **S710**. Alternatively, if the dispenser quality is unsatisfactory, no in step S**708**, the white pixel waveform is loaded and maintenance is initiated again in step S**704**.

The dispenser waveform can be changed from single primary drop regime to nozzle wetting with drop formation regime by shifting the basic jetting parameters. During maintenance activities, each nozzle can be kept wetted while removing the accumulated resist which increases the viable process window for dispensers with and without a non-wetting coating on the faceplate. The dispenser waveform is changed by shifting the basic jetting parameters such that each nozzle can be kept wetted while removing accumulated resist during maintenance of the dispenser.

Further modifications and alternative embodiments of various aspects will be apparent to those skilled in the art in view of this description. Accordingly, this description is to be construed as illustrative only. It is to be understood that the forms shown and described herein are to be taken as examples of embodiments. Elements and materials may be substituted for those illustrated and described herein, parts and processes may be reversed, and certain features may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of this description.

What is claimed is:

1. A method of cleaning a dispenser including a plurality of nozzles with a faceplate and a plurality of jetting units for jetting a material from the plurality of nozzles, wherein each of the plurality of nozzles has one of the plurality of jetting units, the method comprising:

applying a suction force onto a surface of the faceplate using a suction apparatus;

translating the suction apparatus relative to the faceplate such that a portion of nozzles from the plurality of nozzles being exposed to the suction force changes; and

vibrating each meniscus of the portion of nozzles being exposed to the suction force by the suction apparatus by operating the jetting units,

wherein the suction force is applied without the surface and the suction apparatus being in contact, and

wherein, in the vibrating, each meniscus associated with nozzles from the plurality of nozzles that are not being exposed to the suction force is not vibrated,

wherein the suction apparatus includes a rectangular orifice with lips that surround the rectangular orifice, the suction apparatus includes a characteristic width, which is a sum of a narrowest width of the rectangular orifice and twice a width of the lips,

wherein, in the vibrating, the characteristic width is less than an excited width of the dispenser, the excited width referring to a width of the portion of nozzles with their associated menisci being vibrated while being exposed to the suction force.

2. The method according to claim **1**, wherein a number of nozzles associated with the portion of nozzles being exposed to the suction force is less than a total number of the plurality of nozzles.

3. The method according to claim **1**, wherein the faceplate has a non-wetting coating.

4. The method according to claim **1**, wherein the dispenser is a piezoelectric jetting type dispenser.

5. The method according to claim **1**, wherein, in the vibrating, each meniscus being exposed to the suction force by the suction apparatus is vibrated by applying a voltage to the jetting unit for each nozzle that is exposed to the suction force.

6. The method according to claim **5**, wherein, in the vibrating, the voltage applied to the jetting units corresponding to the portion of nozzles is the minimum amount of voltage required to vibrate each meniscus without causing the material to be jetted from the portion of nozzles.

7. The method according to claim **1**, wherein, in the vibrating, the portion of nozzles being exposed to the suction force by the suction apparatus is vibrated so that the portion of nozzles moves in a direction perpendicular to a dispensing direction.

8. The method according to claim **1**, wherein, in the vibrating, a non-jetting waveform is applied to the jetting units corresponding to the portion of nozzles being exposed to the suction force by the suction apparatus.

9. The method according to claim **1**, wherein, in the vibrating, no material is jetted into a process module from the portion of nozzles being exposed to the suction force by the suction apparatus.

10. The method according to claim **1**, wherein each nozzle from the plurality of nozzles forms a meniscus, wherein the meniscus is a curved upper surface of the material in the nozzle.

11. The method according to claim **1**, wherein, in the vibrating, each meniscus of the portion of nozzles is vibrated while the portion of nozzles is exposed to the suction force by the suction apparatus, so that a force applied by the jetting units to the material in the exposed portion of nozzles is below a threshold required to jet the material from the exposed portion of nozzles.

12. The method according to claim **1**, wherein the narrowest width of the rectangular orifice is 0.5 mm.

13. A dispensing system, comprising:

a dispenser configured to jet a material, including a faceplate with a plurality of nozzles and a plurality of jetting units for jetting the material from the plurality of nozzles, wherein each of the plurality of nozzles has one of the plurality of jetting units;

a suction apparatus for applying a suction force onto the faceplate, wherein the suction apparatus includes a rectangular orifice with lips that surround the rectangular orifice, the suction apparatus includes a characteristic width, which is a sum of a narrowest width of the rectangular orifice and twice a width of the lips;

one or more processors; and

one or more memories storing instructions, when executed by the one or more processors, causes the dispensing system to:

applying the suction force onto a surface of the faceplate using the suction apparatus;

translating the suction apparatus relative to the faceplate such that a portion of nozzles from the plurality of nozzles being exposed to the suction force changes; and

vibrating each meniscus of the portion of nozzles being exposed to the suction force by the suction apparatus by operating the jetting units,

wherein the suction force is applied without the surface and the suction apparatus being in contact, and

wherein, in the vibrating, each meniscus associated with nozzles from the plurality of nozzles that are not being exposed to the suction force is not vibrated, and the characteristic width is less than an excited width of the dispenser, the excited width referring to a width of the portion of nozzles with their associated menisci being vibrated while being exposed to the suction force.

14. The dispensing system according to claim **13**, wherein the narrowest width of the rectangular orifice is 0.5 mm.

15. A method of making an article, comprising:

cleaning a dispenser including a faceplate with a plurality of nozzles and a plurality of jetting units for jetting a material from the plurality of nozzles, wherein each of the plurality of nozzles has one of the plurality of jetting units, the cleaning including:

applying a suction force onto a surface of the faceplate using a suction apparatus;

translating the suction apparatus relative to the faceplate such that a portion of nozzles from the plurality of nozzles being exposed to the suction force changes; and

vibrating each meniscus of the portion of nozzles being exposed to the suction force by the suction apparatus by operating the jetting units, wherein the suction force is applied without the surface and the suction apparatus being in contact;

jetting the material onto a substrate using the dispenser;

forming a pattern or a layer of the jetted material on the substrate; and

processing the formed pattern or layer to make the article,

wherein, in the vibrating, each meniscus associated with nozzles from the plurality of nozzles that are not being exposed to the suction force is not vibrated,

wherein the suction apparatus includes a rectangular orifice with lips that surround the rectangular orifice, the suction apparatus includes a characteristic width, which is a sum of a narrowest width of the rectangular orifice and twice a width of the lips,

wherein, in the vibrating, the characteristic width is less than an excited width of the dispenser the excited width referring to a width of the portion of nozzles with their associated menisci being vibrated while being exposed to the suction force.

16. The method of making an article according to claim **15**, wherein the narrowest width of the rectangular orifice is 0.5 mm.

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